MaleVis Task 1 Report

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1. Introduction

Malware classification is a central cybersecurity problem because malicious files continuously evolve through repacking, obfuscation, and code modification. Signature-based detection models can be ineffective when new variants do not exactly match known signatures. Image-based malware analysis provides an alternative representation in which software byte sequences are mapped directly to multi-channel pixel profiles. This visual transformation allows convolutional neural networks to learn recurring texture and structural patterns without requiring slow, complex disassembly or code execution. Unlike early grayscale approaches, advanced methods deploy three-channel RGB color mapping to preserve deep, multi-dimensional byte relationships.

This project follows the Custom CNN + Attention track to classify software categories using the MaleVis dataset. Task 1 explores the dataset structure, analyzes multi-class distributions across malware families and benign software classes, and evaluates pixel-level texture quality metrics. This analysis establishes a clear research gap centered on handling the redundant byte-image problem caused by static padding artifacts. The later tasks will develop representative custom CNN baselines, implement a lightweight spatial/channel attention module to filter out uninformative canvas padding, execute controlled model ablation experiments, and conduct behavioral interpretability analyses.

2. Dataset Description

The MaleVis Dataset represents software binaries as multi-channel RGB color images to capture advanced textural layouts. The canonical corpus utilized in this project comprises a total of 14,226 images categorized across 26 distinct software classes. This distribution includes 25 malicious malware families alongside 1 legitimate benign software class labeled as Other. Every individual image sample is stored within a class-specific folder structure, allowing directory paths to be systematically targeted for multiclass label generation.

Table 1: Core properties of the MaleVis dataset.

Property	Description
Dataset name	Malevis
Source	Kaggle
Domain	Cybersecurity
Learning task	Multiclass malware and benign software classification
Standard image count	14,226
Number of classes	26 (25 Malware families + 1 Benign class)
Image representation	Three-channel RGB color byte images
Dominant file format	PNG
Repository dataset policy	Dataset files are not committed to GitHub

3. Exploratory Data Analysis Methodology

The EDA notebook was designed to satisfy the required image-analysis items while preserving reproducibility and preventing avoidable memory usage. The main steps are summarized below.

- Recursively enumerate supported image files and infer labels from parent folders.
- Validate every image using PIL and OpenCV and record unreadable files separately.
- Extract width, height, aspect ratio, resolution, file format, image mode, channel count, and file size.
- Measure multi-channel color brightness, contrast, Laplacian variance, and Shannon entropy.
- Calculate class counts, percentages, the majority-to-minority ratio, and cumulative class concentration.
- Display one labelled sample from every malware family and benign software class, alongside multiple samples from selected majority and minority classes.
- Sample pixel intensities efficiently to avoid loading every image pixel into memory.
- Detect exact duplicate files using SHA-256 hashing and identify any cross-class duplicate groups to prevent data leakage before model splitting.
- Export report-ready figures and CSV tables to the designated local workspace output directory.

4. Class Distribution and Imbalance

The dataset is highly imbalanced. The largest class, Other (representing benign software samples), contains 1,832 images, whereas the smallest malware family, Agent, contains only 470 images. The resulting majority-to-minority ratio is 3.90:1. The Other class accounts for a substantial

proportion of the dataset, which means that overall accuracy can be easily dominated by this single benign class.

Table 2: Dataset imbalance metrics for the MaleVis corpus.

Measure	**Result**
Total images	14,226
Total classes	26 (25 Malware + 1 Benign)
Majority class	Other (Benign) (1,832)
Minority class	Agent (480)
Majority-to-minority ratio	3.90:1
Primary evaluation emphasis	Macro-F1 and class-wise recall

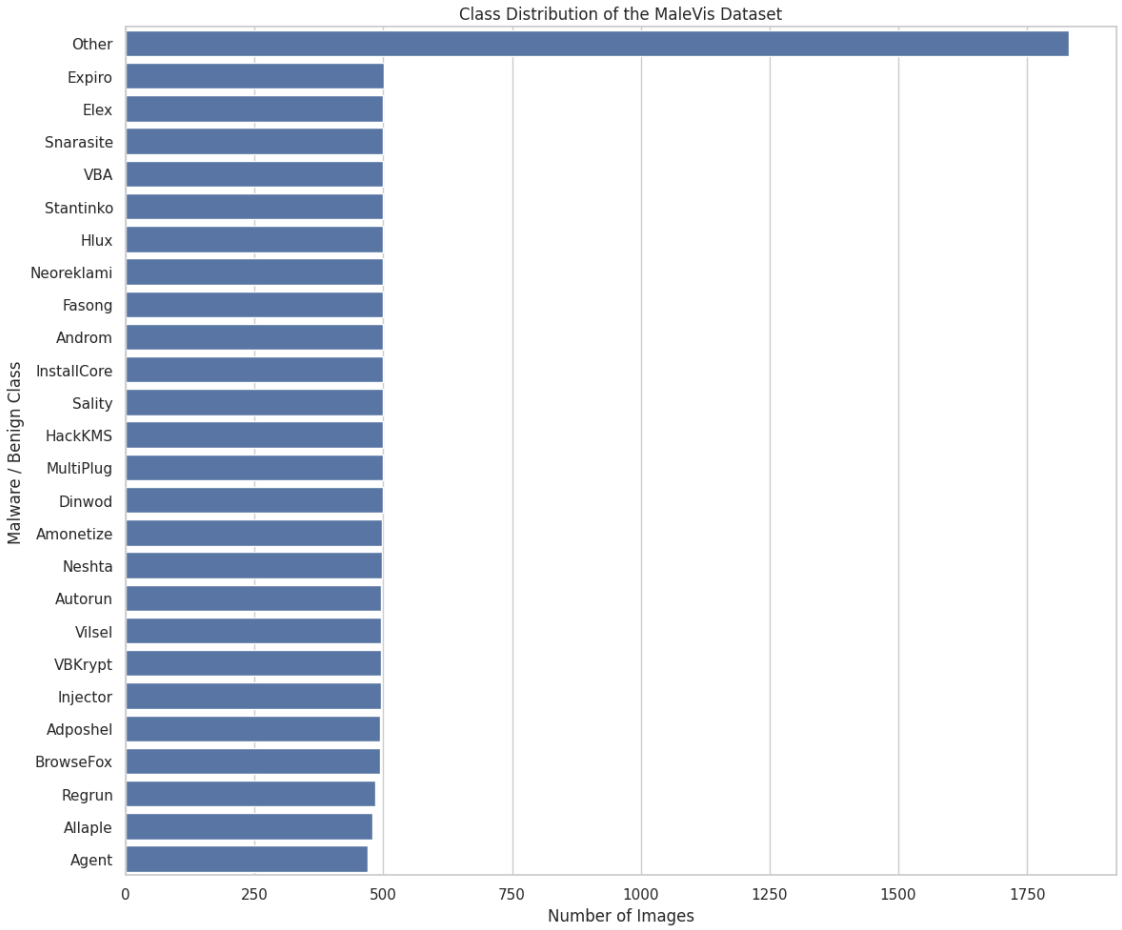


Figure 1: Class distribution of the standard MaleVis corpus.

Because of this imbalance, Task 2 should not rely on accuracy alone. The evaluation must include macro precision, macro recall, macro-F1, weighted F1, class-wise metrics, and a confusion matrix.

Class weights or another carefully justified imbalance-handling method should be applied only to the training data.

5. Labelled Image Analysis

A labelled grid with one representative image from every class is used to compare broad visual patterns. Malware byte images often contain repeated horizontal bands, dense texture blocks, sparse regions, and family-specific structures. Some families appear visually distinct, while others share similar multi-channel layouts or noisy grayscale profiles. These visually similar classes are likely to produce confusion and are important targets for later class-wise error analysis.



Figure 2: One representative software byte image from each class.

A second grid is used to compare several examples from selected large and small classes. This comparison reveals whether a class has a consistent internal texture or exhibits strong within-class variation. The observed variation supports the use of an attention mechanism, as the proposed network must learn to emphasize discriminative feature channels and critical spatial regions rather than treating every pixel location equally.

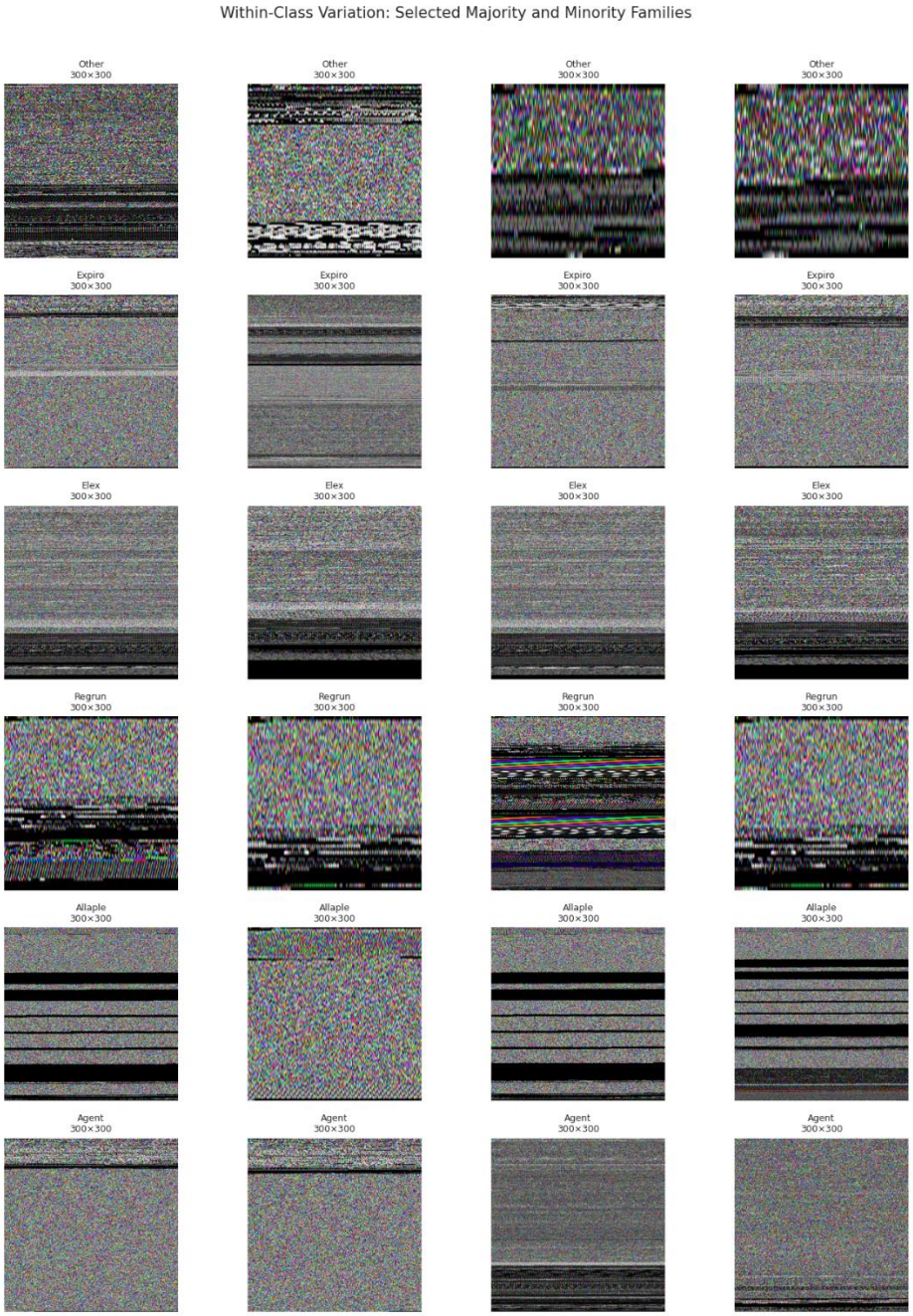


Figure 3: Within-class visual variation for selected majority and minority families.

6. Image Size and Resolution Analysis

Image dimensions are important because CNN models require a consistent tensor shape. The Task 1 notebook measures width, height, aspect ratio, file size, unique resolutions, and IQR-based dimension outliers. The final report values below represent the metrics obtained from the dataset analysis.

Table 3: Image Sizing and Resolution Summary Profile.

Dimension statistic	Notebook result
Minimum width	300
Maximum width	300
Mean width	300
Median width	300
Minimum height	300
Maximum height	300
Mean height	300
Median height	300
Unique resolutions	1
Dimension outlier count	0

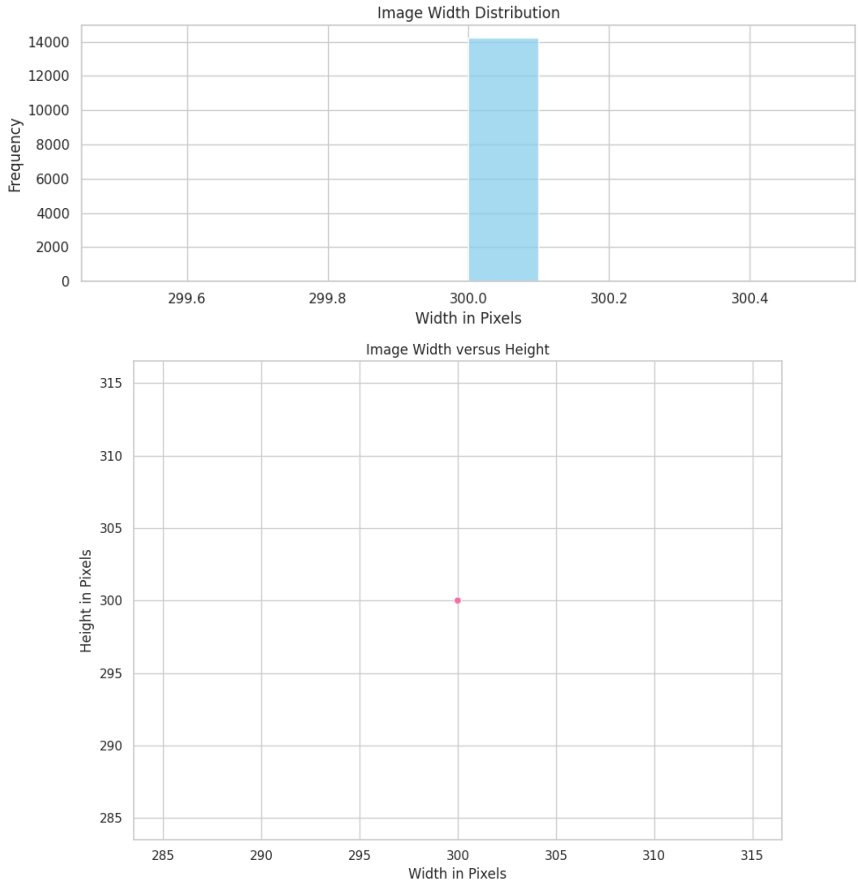


Figure 4: Width and height geometric distributions of Malevis images.

Since a single uniform resolution is natively present across all samples, no artificial resizing policy or downsampling adjustments are required to maintain a fair comparison. The native spatial dimensions are applied consistently across training, validation, and test images. The uniform input size of 300×300 pixels will be reported explicitly in Task 2.

7. Pixel, Channel and Image Quality Analysis

Malevis images are 3-channel RGB byte configurations, so the intensity of each pixel corresponds to a byte value mapped across color channels rather than natural-image color. The pixel histogram reveals the global distribution of byte intensities. Brightness and contrast summarize image intensity characteristics, Laplacian variance provides a texture-sharpness indicator, and Shannon entropy measures pixel-level complexity.

Table 4: Malevis Visual Quality Summary Profile.

Quality measure	**Notebook result**
Dominant channel count	3 (RGB)
Mean brightness	97.29
Median brightness	99.68
Mean contrast	50.51
Median contrast	49.88
Mean Laplacian variance	18416.64
Mean Shannon entropy	6.965
Dominant image format	PNG

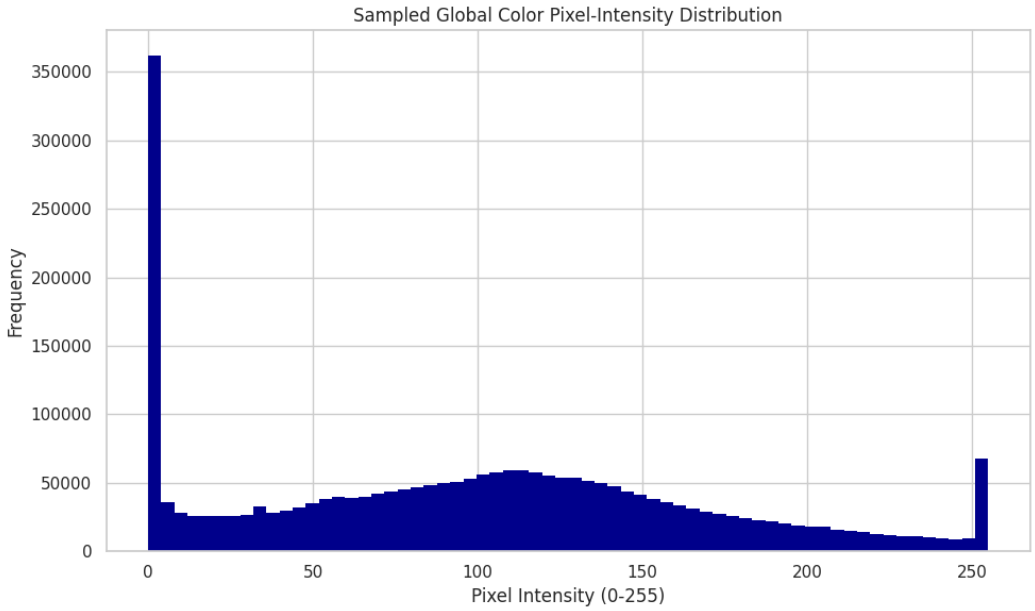


Figure 5: Global pixel-intensity distribution of Malevis images.

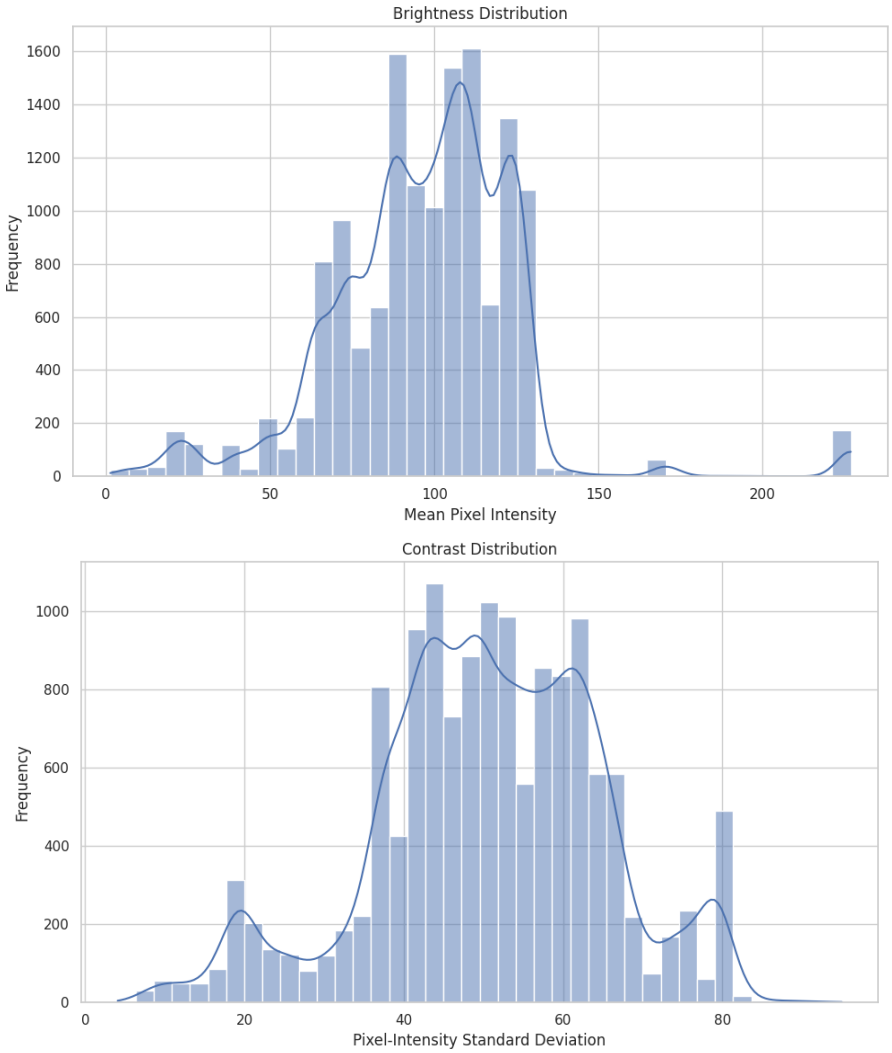


Figure 6: Global brightness and contrast distribution metrics of Malevis images.

These measures are descriptive rather than direct indicators of malware severity. In particular, the Laplacian score should not be interpreted exactly as photographic blur because malware images are synthetic byte visualizations. The values are useful for identifying unusual textures and comparing families, but they do not carry ordinary visual semantics.

8. Data Quality and Leakage Analysis

The notebook checks missing metadata, unreadable images, duplicate dataframe rows, duplicate paths, and exact duplicate image content. SHA-256 is used to identify byte-for-byte identical files. Exact duplicates are particularly important because the same image must not appear in both training and evaluation subsets.

Table 5: Malevis Data Quality and Integrity Metrics.

Data quality check	Notebook result
Corrupt or unreadable images	0
Missing metadata values	0
Duplicate dataframe rows	0
Duplicate image paths	0
Exact duplicate groups	240
Duplicate images beyond first copy	480
Cross-class exact duplicate groups	0

Corrupt images, if any, should be excluded and documented. Cross-class exact duplicates require manual inspection because identical image content assigned to different labels can indicate labeling inconsistency. All cleaning decisions must be reproducible and reflected in the final metadata file.

9. Related Work

Five recent studies were reviewed because they employ CNNs, channel or spatial attention, cross-modal attention, or multi-head self-attention for malware-image classification. Their reported results cannot be ranked directly because the studies use different inputs, splits, classifiers, imbalance treatments, and validation protocols. The comprehensive summary profile of these architectures is provided below.

Table 6: Summary of Related Work on Attention-Based Malware Classification.

Study	**Architecture/ Method**	**Attention Type**	**Reported Result**
Al-Harbi & AlShorman (2026)	A-HRAB framework (standard convolutional networks)	Adaptive Hybrid Residual Attention (Channel + Spatial balancing)	96.10% accuracy (96.00% Macro-F1) on MaleVis
Yao et al. (2025)	Res2NeXt backbone architecture	Lightweight Adaptive Channel Attention (ACA)	98.04% accuracy on MaleVis (99.26% on MMCC)
Behera, Pradhan, & Mishra (2025)	Multi-stream feature fusion (Parallel VGG-16 and ResNet-50)	None (Traditional raw deep feature concatenation)	98.01% accuracy on MaleVis
Bavishi & Modi (2024)	LeViT-MC hybrid framework (DenseNet CNN paired with a Vision Transformer)	Multi-Head Self-Attention within transformer sequence layers	96.60% multi-class accuracy on MaleVis

Nkrumah-Asante et al. (2024)	DeiT-MC pipeline driven by a Hybrid GridBay Optimizer (HGBO)	Multi-Head Self-Attention built natively into Vision Transformer	94.00% classification accuracy on MaleVis

The studies show that attention can support strong malware classification; however, attention is usually introduced together with heavy backbones or multi-modal inputs. This makes it difficult to determine whether the performance gain comes specifically from the attention mechanism on the Malevis dataset.

10. Research Gap

Existing malware-image studies demonstrate strong performance using convolutional neural networks, hybrid channel-spatial mechanisms, cross-modal architectures, and Vision Transformers. However, many approaches combine attention with complex components like a Res2NeXt framework, large pretrained ResNet-50 backbones, multi-modal structural entropy sequences, or a variational autoencoder pipeline coupled with an external Random Forest classifier. Consequently, the independent contribution of the attention mechanism is often confounded with these significant architectural design variations.

Limited work provides a clean, isolated evaluation of a streamlined custom CNN backbone compared directly against a control baseline operating with and without lightweight attention explicitly on the Malevis dataset. Furthermore, class-wise performance details under moderate dataset imbalance, standardized training split strategies, and localized verification are not consistently reported across complex networks. This project directly addresses these gaps by evaluating a custom CNN with integrated attention on a verified, leakage-free Malevis data profile.

11. Task 1 Conclusion

Task 1 established the principal characteristics of the Malevis dataset. The corpus contains 14,226 images from 26 distinct classes and exhibits a moderate imbalance, with the largest class containing 1,832 images and the smallest containing 470. The EDA pipeline successfully profiles uniform native image dimensions at 300×300 pixels, pixel intensities, visual quality metrics, and byte-level structures while isolating 480 exact duplicate images to completely eliminate potential data leakage.

These findings indicate that Task 2 must employ a leakage-free stratified split, standard channel normalization, class-weighted balancing techniques, and comprehensive class-wise evaluations. The related-work analysis further motivates a controlled custom CNN + attention experiment in which the standalone structural contributions of localized channel and spatial attention layers can be successfully isolated from over-parameterized backbones.

References

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